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Topic :- Linux Commands

STEP 1 :- Update the ubuntu once installed use command 'apt-get update -y'

```
root@ip-172-31-80-49: ~  
root@ip-172-31-80-49:~# pwd  
/root  
root@ip-172-31-80-49:~# apt-get update -y  
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease  
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]  
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]  
Get:4 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]  
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]  
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]  
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]  
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]  
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]  
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]  
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2598 kB]  
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [421 kB]  
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [18.5 kB]  
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [3585 kB]  
Get:15 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2354 kB]  
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [640 kB]  
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 c-n-f Metadata [676 B]  
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1208 kB]  
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [298 kB]  
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [28.7 kB]  
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [46.5 kB]  
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/multiverse Translation-en [11.8 kB]  
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [592 B]  
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [68.8 kB]  
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/main Translation-en [11.4 kB]  
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [392 B]  
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 c-n-f Metadata [116 B]  
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [30.0 kB]  
Get:29 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/universe Translation-en [16.5 kB]  
Get:30 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 c-n-f Metadata [672 B]  
Get:31 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 c-n-f Metadata [116 B]  
Get:32 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [357 kB]  
Get:33 http://security.ubuntu.com/ubuntu jammy-security/main amd64 c-n-f Metadata [13.6 kB]  
Get:34 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [3466 kB]  
Get:35 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [620 kB]  
Get:36 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 c-n-f Metadata [624 B]  
Get:37 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [974 kB]  
Get:38 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [210 kB]  
Get:39 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [21.7 kB]  
Get:40 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [39.6 kB]  
Get:41 http://security.ubuntu.com/ubuntu jammy-security/multiverse Translation-en [8716 B]  
Get:42 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 c-n-f Metadata [368 B]  
Fetched 37.8 MB in 19s (2022 kB/s)  
Reading package lists... Done  
root@ip-172-31-80-49:~# |
```

**1 :- Commands used Head and Tail ; Head -2 Tail -2 ;
Head -n2 ; Tail -n2.**

```
root@ip-172-31-91-254: ~/AWS
root@ip-172-31-91-254:~# mkdir -p AWS/Devops/Cloud/New
root@ip-172-31-91-254:~# ls
AWS  snap
root@ip-172-31-91-254:~# cd AWS
root@ip-172-31-91-254:~/AWS# touch file.txt
root@ip-172-31-91-254:~/AWS# vi file.txt
root@ip-172-31-91-254:~/AWS# head file.txt
Hello All
one
two
three
four
five
six
seven
eight
nine
root@ip-172-31-91-254:~/AWS# tail file.txt
fourteen
sixteen
seventeen
eighteen
nineteen
twenty
twenty one
twenty two
twenty three
twenty four
root@ip-172-31-91-254:~/AWS# |
```

2 :- Commands used ssh-keygen vi id_rsa; vi id_rsa.pub;

cat id_rsa; cat_rsa.pub

```
root@ip-172-31-91-254: ~/.ssh
root@ip-172-31-91-254:~/AWS# cd
root@ip-172-31-91-254:~# cd .ssh
root@ip-172-31-91-254:~/.ssh# ls
authorized_keys
root@ip-172-31-91-254:~/.ssh# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:Vvpb2U13rIYClABVX6b8B/EwA8FBQVzsbMgvNOxeJjM root@ip-172-31-91-254
The key's randomart image is:
+---[RSA 3072]-----+
|      .oo..=BBO      |
|      . +o+. *       |
|      oo=+. .        |
|      . O*.+. .      |
|      So +. . =      |
|      . OE += +o     |
|      .OB+ + .       |
|      .+ .           |
|      .              |
+-----[SHA256]-----+
root@ip-172-31-91-254:~/.ssh# ls
authorized_keys id_rsa id_rsa.pub
root@ip-172-31-91-254:~/.ssh# |
```

```
root@ip-172-31-91-254: ~/AWS
root@ip-172-31-91-254:~/AWS# cat file.txt
Hello All
one
two
three
four
five
six
seven
eight
nine
ten
eleven
twelve
thirteen
fourteen
sixteen
seventeen
eighteen
nineteen
twenty
twenty one
twenty two
twenty three
twenty four
root@ip-172-31-91-254:~/AWS# head -2 file.txt
Hello All
one
root@ip-172-31-91-254:~/AWS# head -n2 file.txt
Hello All
one
root@ip-172-31-91-254:~/AWS# tail -2 file.txt
twenty three
twenty four
root@ip-172-31-91-254:~/AWS# tail -n2 file.txt
twenty three
twenty four
root@ip-172-31-91-254:~/AWS# |
```

root@ip-172-31-91-254: ~/.ssh

```
-----BEGIN OPENSSH PRIVATE KEY-----
b3B1bnNzac1rZxktjdjEAAAAAG5vbmUAAAAEbm9uZQAAAAAAAAABAAABlwAAAdzc2gtcn
NhAAAAAwEAAQAAAYEaxb/vvaji38e80a5Mnypwo5/wm9PWza7eZYLknwnAjcecur/w4xxvz
6EWq2360Xon85kUgzzf7Isk4tpjed7Aykv10011tfxkxer3xw7cUvA+7VIOjY0wmnEka4O
D/Xglvhs5+9Thakd01NvsQPC4B2P2XitcvNGkyfP1wWjX5m/MqC+Ztas1pymmrjEwhi1nN
f+v+94RS01cxTubFU9JcewsdeTBqozHg/jkcyMhsn45vur952aFoZsmhiA2B8+UsH9wc90
pirmKIPZZ05L46ntudlp6zb1lvWDRz6w/crNzLEeDSiv66He+omdjz6v4mPtG4DSBjnmOM
FM5wckN/ZVltoSts4n06eT3dCD9+oBc5ynV+Z2aupF+VTFtus4Nzh/oQU5INruEa167weF
1xzok2cvwqpX3j6dTVq1qyfnucwK/va/zTF/03LqfhsNaAdOTmsbhcicQs7oo4i51cxoBd
S9xm1/wv8Fc8UxafhinTZ1jJy51B92XZfCuN/bg1AAAFkFnmGjXZ5oCccAAAAB3Nzac1yc2
EAAAGBAMw/1b2ot/HvNGuTj8qcdUf8JvT1s2u3mWC5j1pwI3HhLq/1uMcb8+hFqtt+j1zp
/OS1IM2X+yLJOCKY3newMpFdndJdbx8ZMXq98vu3FFWvu1SNCWNfppxCmuBg/14jb4bofV
U4WindjTVbEDwuAdj914rXLzRisnz9cFicezvzKgvmbwrNacppq4xMIYtZZX/lfveEutCX
MU7mxVPSXh1rA3kwaJmR4P45HGDIup+ob7kfedmhaGupoYgNgfP1LB/VnPdKyq5i1d2Wdo
s+op01Hzaes24tb1g0c+lv3KzcyxHg0or+uh3vqDHY2er+jj7RuA0gsZzNDBTOVncp/2VZ
baErUuJ9onk93og/fqAXocplfmdm1krf1uxbvlODwyfzkFOSda7hgteu8Hn5v2TpNnL1qq
V9yenu1atasnzbFiv72v803/9Ny0H4bdWgHTkzLG4XiGkLozjucodxMTg00vcZtf1lfBX
PFF2n4Yp02dSScudQfd12xwrjf24jQAAAAAMBAEAAAGACH2qmdCSxwkd/5h79ukogrGDAR
wiFjF3j/2jQpDbtmJwcNu+KV575jJNM+EUV0CrNaIR/Gkqoo281VZCA6avT8kcVXT/Ucz
psmNGYEM8HjxNDoyEv8knkUF3w4LYP/iuAkZznGy/WUvedEFK3BWR/jrjtyafEdynqFLN6
LQXe/c+FWHlOvpC3XqFCyXEqkyfjFeeEZIQaId52N1saoyqmX76jI3/wzUG1Zdb5jqact1
5j8DXmLd1qoo82yRDEXjIbzIpOCZMmIxc8SMD0YB405PC0BCQ+pmcdhRNMf+ex/nfuFB2U
1IWZL+Sw8yZzrr+ywneUrmCtu2DQYFaLNaGK64ELtezQkvyeA6hyod1FgFjw4oqM+i0YK
5tEQI3rBZKFkFmhuEzInokhyFSud50XwgZqWZZ0XnpY8QrvjHjuy1BcscbFh1w1p/LFpFH
q7RFZzNmEN849TTbbdc5IfHDOit0jbnMTs1LSGwLm71Hxwhb9WdnQzWEXmrZz1G2LBAAAA
wQC8+p/k5kQ3ryEnZ2Sr2s2giGxn3mtP5cIZnrV/H3A87ziep/efBLT7f1FdPktzpdQn/q
86p2HhvvWhZMigrXksGHnKwM4ZXf7/HeqMKASRTSMQ0r9wy50FFG1it26sdFccSQwOCiZHX
0fxbu9FmthOHrUwFqrj7Qmh6gULMbJyzTmkv+9a0B8b1dCT1tptry3pkjhxof/2qhJWnyDU
I4aRBW7tszN2jn+5mPlkRkm7yeewev6naPMG/OWuntbMf5ZTWAAADBApXuzRIBqLluxdXP
SFGrvQtd7kViyhASUNG9P1MG+0BG9NBVHrKu8Zs+wNCT+cQin0PPw77fx5o+Hofj/fsgbp
Xgyz51ug8jmZCxpSvntyrNV7Qo6GdS9wsUR1n+VZiuywie5vt4knq17DhpdRZrb/ez4AkH
hCm/VGNWtZzXC9fyowbhroHBawM/MnFxmHy8ML4d6b0vg5aF+2p1c7WY/34PyH/YnQvm+
Xvm54bh64vonB/EDVzcExa6fehYETZZQAAAMEazdhyJk7n4nou8DFx3gexozc4oxfpwgrk
pKukVvuzDh7AtunnsQqf8wop92WLDXKoYo5t8T6b+gd58o193+hB1FdFGLWuJganXv5WjJ
qEUGPSjI1liRgokQDH9Iple4uFzqptwgfjoPv8Lt5SEeusNntbt57a8pojrqiCBKAmGrng
mI+J6a1FHV39aguLCKyi4lJQU7N1sdIw7Bt/ct/WLXTxKEPF1OWCrj0Z8hXDU381oowypJ
w7bw5f02Znrp+5AAAAFXJvb3RAaXAtMTcyLTMTkxLT1NAECAWQF
-----END OPENSSH PRIVATE KEY-----
"id_rsa" 38L, 2610B
```

root@ip-172-31-91-254: ~/.ssh

```
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDQFv9W9qLfx7zRrkyfKna7n/Cb09bnrt51gusdaccnx5y6v9bjHG/PoRarbf05c6FzKpSDNl/siyTgimN53sDKRrXTSxw1/GTF
6vFFbtXsvr7tUjQ1jRaacQprq499ecwGzn7lOPop35u1wxA8Lghy/zeKi180Yrj8/XBYnHmb8yol5m1qzwnkaauMTcGLwC1/5x73hFLQ1zFo3svT01x5awN5Mgo5ked+ORXgyf
Kfjm+5H3nZowh1kaGIDYHz3SwfLz3Smkuyog91ntkvjgdR2nnuLU9YNHp9y53msr4KK/rod76gx2Nnq/1Y40bgN1EmczQwuz1Zwqf91Ww2hk1L1fTp5Pd0rp36gfZnknd
X5nzpskx5VMW1szg1mH85BTkg2u4RrXrv85+Vdk6Tzy9aqlfCnp1Nwrwr3825xyr+9r/NN//TctB+Gw1oB05MyxuFyIJCzs47gjnVZE4ENL3GbX9ZxwVzxrDp+GkdNnuKLnUH3
zd18k439ucu= root@ip-172-31-91-254
```

"id_rsa.pub" 1L, 575B

1,1

A11

3 :- Commands used wc file name; wc -c file name; wc -l file name; wc -w file name

```
root@ip-172-31-91-254: ~/AWS
root@ip-172-31-91-254:~/AWS# cat file.txt
Hello All
one
two
three
four
five
six
seven
eight
nine
ten
eleven
twelve
thirteen
fourteen
sixteen
seventeen
eighteen
nineteen
twenty
twenty one
twenty two
twenty three
twenty four
root@ip-172-31-91-254:~/AWS# wc file.txt
 24  29 182 file.txt
root@ip-172-31-91-254:~/AWS# wc -l file.txt
24 file.txt
root@ip-172-31-91-254:~/AWS# wc -c file.txt
182 file.txt
root@ip-172-31-91-254:~/AWS# wc -w file.txt
29 file.txt
root@ip-172-31-91-254:~/AWS# |
```

4 :- Commands used lsblk history

```
root@ip-172-31-91-254: ~
root@ip-172-31-91-254:~/AWS# cd
root@ip-172-31-91-254:~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0        7:0      0 26.3M  1 loop /snap/amazon-ssm-agent/9881
loop1        7:1      0 63.7M  1 loop /snap/core20/2496
loop2        7:2      0 89.4M  1 loop /snap/lxd/31333
loop3        7:3      0 73.9M  1 loop /snap/core22/1748
loop4        7:4      0 44.4M  1 loop /snap/snapd/23545
xvda         202:0     0   8G   0 disk
├─xvda1      202:1     0  7.9G   0 part /
├─xvda14     202:14    0   4M   0 part
└─xvda15     202:15    0 106M   0 part /boot/efi
root@ip-172-31-91-254:~# history
  1 cd
  2 clear
  3 apt get update-y
  4 apt-getupdate -y
  5 apt-get update
  6 clear
  7 mkdir -p AWS/Devops/Cloud/New
  8 ls
  9 cd AWS
 10 touch file.txt
 11 vi file.txt
 12 head file.txt
 13 tail file.txt
 14 head -3 file.txt
 15 head -2 file.txt
 16 tail -2 file.txt
 17 clear
 18 cat file.txt
 19 head -2 file.txt
 20 head -n2 file.txt
 21 tail -2 file.txt
 22 tail -n2 file.txt
 23 clear
 24 cd
 25 cd .ssh
 26 ls
```


5 :- Commands used cal

```
root@ip-172-31-91-254:~# apt install ncal
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  ncal
0 upgraded, 1 newly installed, 0 to remove and 91 not upgraded.
Need to get 20.2 kB of archives.
After this operation, 69.6 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 ncal amd64 12.1.7+nmu3ubuntu2 [20.2 kB]
Fetched 20.2 kB in 0s (1025 kB/s)
Selecting previously unselected package ncal.
(Reading database ... 65857 files and directories currently installed.)
Preparing to unpack .../ncal_12.1.7+nmu3ubuntu2_amd64.deb ...
Unpacking ncal (12.1.7+nmu3ubuntu2) ...
Setting up ncal (12.1.7+nmu3ubuntu2) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-91-254:~# cal june 2025
      June 2025
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30
root@ip-172-31-91-254:~#
```

6 :- Commands used apt install apache2; apache2 -version; systemctl status apache2; systemctl stop apache2; systemctl start apache2 ; systemctl enable apache2

```
root@ip-172-31-91-254:~# apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0 mailcap
  mime-support ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser bzip2-doc
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0
  mailcap mime-support ssl-cert
0 upgraded, 13 newly installed, 0 to remove and 91 not upgraded.
Need to get 2142 kB of archives.
After this operation, 8528 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [108 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [92.8 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9170 B]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 liblua5.3-0 amd64 5.3.6-1build1 [140 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.14 [1349 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.14 [165 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.14 [89.0 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 mailcap all 3.70+nmu1ubuntu1 [23.8 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 mime-support all 3.66 [3696 B]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2 amd64 2.4.52-1ubuntu4.14 [97.9 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 bzip2 amd64 1.0.8-5build1 [34.8 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 ssl-cert all 1.1.2 [17.4 kB]
```

```

root@ip-172-31-91-254: ~
root@ip-172-31-91-254:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-06-03 10:10:58 UTC; 7min ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 2531 (apache2)
      Tasks: 55 (limit: 1129)
    Memory: 5.6M
      CPU: 50ms
    CGroup: /system.slice/apache2.service
            └─2531 /usr/sbin/apache2 -k start
              └─2533 /usr/sbin/apache2 -k start
                └─2534 /usr/sbin/apache2 -k start

Jun 03 10:10:58 ip-172-31-91-254 systemd[1]: Starting The Apache HTTP Server...
Jun 03 10:10:58 ip-172-31-91-254 systemd[1]: Started The Apache HTTP Server.
root@ip-172-31-91-254:~# systemctl stop apache2
root@ip-172-31-91-254:~# systemctl status apache2
○ apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: inactive (dead) since Tue 2025-06-03 10:18:38 UTC; 32s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Process: 2877 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
    Main PID: 2531 (code=exited, status=0/SUCCESS)
      CPU: 69ms

Jun 03 10:10:58 ip-172-31-91-254 systemd[1]: Starting The Apache HTTP Server...
Jun 03 10:10:58 ip-172-31-91-254 systemd[1]: Started The Apache HTTP Server.
Jun 03 10:18:38 ip-172-31-91-254 systemd[1]: Stopping The Apache HTTP Server...
Jun 03 10:18:38 ip-172-31-91-254 systemd[1]: apache2.service: Deactivated successfully.
Jun 03 10:18:38 ip-172-31-91-254 systemd[1]: Stopped The Apache HTTP Server.
root@ip-172-31-91-254:~# systemctl start apache2
root@ip-172-31-91-254:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-06-03 10:19:27 UTC; 35s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Process: 2887 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)

root@ip-172-31-91-254: ~
root@ip-172-31-91-254:~# systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable apache2
root@ip-172-31-91-254:~#

```

7 :- Commands used useradd swapnil; passwd swapnil; su swapnil in this case as we did not fire command properly while adding user permissions would be denied no command will run.

```

root@ip-172-31-91-254: ~
root@ip-172-31-91-254:~# systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable apache2
root@ip-172-31-91-254:~# useradd swapnil
root@ip-172-31-91-254:~# passwd swapnil
New password:
Retype new password:
passwd: password updated successfully
root@ip-172-31-91-254:~# su swapnil
$ ls
ls: cannot open directory '.': Permission denied
$ ls -l
ls: cannot open directory '.': Permission denied
$ exit
root@ip-172-31-91-254:~#

```


8 :- Commands used **useradd -m -d/home/swapnilsanap swapnilsanap; passwd swapnilsanap; cat /etc/passwd**

Visudo = to install packages inside the user we have to permit user first. To permit follow steps

- 1) Go to root
- 2) Enter **visudo**
- 3) Go to user privilege permission
- 4) Add Swapnilsanap ALL=(ALL:ALL) ALL
- 5) Press ctrl x
- 6) Press Y and enter
- 7) Try installing package it will be successful.

root@ip-172-31-91-254: ~

root@ip-172-31-91-254:~# useradd -m -d /home/swapnilsanap

Usage: useradd [options] LOGIN

useradd -D

useradd -D [options]

Options:

--badnames	do not check for bad names
-b, --base-dir BASE_DIR	base directory for the home directory of the new account
--btrfs-subvolume-home	use BTRFS subvolume for home directory
-c, --comment COMMENT	GECOS field of the new account
-d, --home-dir HOME_DIR	home directory of the new account
-D, --defaults	print or change default useradd configuration
-e, --expiredate EXPIRE_DATE	expiration date of the new account
-f, --inactive INACTIVE	password inactivity period of the new account
-g, --gid GROUP	name or ID of the primary group of the new account
-G, --groups GROUPS	list of supplementary groups of the new account
-h, --help	display this help message and exit
-k, --skel SKEL_DIR	use this alternative skeleton directory
-K, --key KEY=VALUE	override /etc/login.defs defaults
-l, --no-log-init	do not add the user to the lastlog and faillog databases
-m, --create-home	create the user's home directory
-M, --no-create-home	do not create the user's home directory
-N, --no-user-group	do not create a group with the same name as the user
-o, --non-unique	allow to create users with duplicate (non-unique) UID
-p, --password PASSWORD	encrypted password of the new account
-r, --system	create a system account
-R, --root CHROOT_DIR	directory to chroot into
-P, --prefix PREFIX_DIR	prefix directory where are located the /etc/* files
-s, --shell SHELL	login shell of the new account
-u, --uid UID	user ID of the new account
-U, --user-group	create a group with the same name as the user
-Z, --selinux-user SEUSER	use a specific SEUSER for the SELinux user mapping
--extrausers	Use the extra users database

root:x:0:0:root:/root:/bin/bash

daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin

bin:x:2:2:bin:/bin:/usr/sbin/nologin

sys:x:3:3:sys:/dev:/usr/sbin/nologin

sync:x:4:65534:sync:/bin:/bin/sync

games:x:5:60:games:/usr/games:/usr/sbin/nologin

man:x:6:12:man:/var/cache/man:/usr/sbin/nologin

lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin

mail:x:8:8:mail:/var/mail:/usr/sbin/nologin

news:x:9:9:news:/var/spool/news:/usr/sbin/nologin

uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin

proxy:x:13:13:proxy:/bin:/usr/sbin/nologin

www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin

backup:x:34:34:backup:/var/backups:/usr/sbin/nologin

list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin

irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin

gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin

nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin

systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin

systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin

messagebus:x:102:105:/nonexistent:/usr/sbin/nologin

systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin

syslog:x:104:111:/home/syslog:/usr/sbin/nologin

_apt:x:105:65534:/nonexistent:/usr/sbin/nologin

tss:x:106:112:TPM software stack,,,:/var/lib/tpm:/bin/false

uidd:x:107:113:/run/uidd:/usr/sbin/nologin

tcpdump:x:108:114:/nonexistent:/usr/sbin/nologin

sshd:x:109:65534:/run/sshd:/usr/sbin/nologin

pollinate:x:110:1:/var/cache/pollinate:/bin/false

landscape:x:111:116:/var/lib/landscape:/usr/sbin/nologin

fwupd-refresh:x:112:117:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin

ec2-instance-connect:x:113:65534:/nonexistent:/usr/sbin/nologin

_chrony:x:114:121:chrony daemon,,,:/var/lib/chrony:/usr/sbin/nologin

ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash

lxd:x:999:100:/var/snap/lxd/common/lxd:/bin/false

swapnil:x:1001:1001:/home/swapnil:/bin/sh

swapnils:x:1002:1002:/home/swapnils:/bin/sh

swapnilss:x:1003:1003:/home/swapnilss:/bin/sh

root@ip-172-31-91-254:~#

```
root@ip-172-31-91-254: ~  
GNU nano 6.2 /etc/sudoers.tmp  
# This file MUST be edited with the 'visudo' command as root.  
#  
# Please consider adding local content in /etc/sudoers.d/ instead of  
# directly modifying this file.  
#  
# See the man page for details on how to write a sudoers file.  
#  
Defaults      env_reset  
Defaults      mail_badpass  
Defaults      secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"  
Defaults      use_pty  
  
# This preserves proxy settings from user environments of root  
# equivalent users (group sudo)  
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"  
  
# This allows running arbitrary commands, but so does ALL, and it means  
# different sudoers have their choice of editor respected.  
#Defaults:%sudo env_keep += "EDITOR"  
  
# Completely harmless preservation of a user preference.  
#Defaults:%sudo env_keep += "GREP_COLOR"  
  
# While you shouldn't normally run git as root, you need to with etckeeper  
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"  
  
# Per-user preferences; root won't have sensible values for them.  
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"  
  
# "sudo scp" or "sudo rsync" should be able to use your SSH agent.  
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"  
  
# Ditto for GPG agent  
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"  
  
AG Help      AQ Write out  AW Where Is  AK Cut       AJ Execute   AL Location  M-U Undo     M-A Set Mark  
AX Exit      AR Read File  AX Replace   AU Paste     AJ Justify   AL Go To Line M-E Redo     M-G Copy
```

```
root@ip-172-31-84-203: ~  
root@ip-172-31-84-203:~# useradd -m -d /home/swapnilsanap swapnilsanap  
root@ip-172-31-84-203:~# passwd  
New password:  
Retype new password:  
Sorry, passwords do not match.  
passwd: Authentication token manipulation error  
passwd: password unchanged  
root@ip-172-31-84-203:~# passwd swapnilsanap  
New password:  
Retype new password:  
passwd: password updated successfully  
root@ip-172-31-84-203:~# visudo  
root@ip-172-31-84-203:~# su - swapnilsanap  
$ apt install tree  
E: Could not open lock file /var/lib/dpkg/lock-frontent - open (13: Permission denied)  
E: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontent), are you root?  
$ exit  
root@ip-172-31-84-203:~# visudo  
root@ip-172-31-84-203:~# su - swapnilsanap  
$ sudo apt-get update  
[sudo] password for swapnilsanap:  
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease  
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]  
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]  
Get:4 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]  
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]  
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]  
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2354 kB]  
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]  
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]  
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]  
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]  
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2619 kB]  
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [423 kB]  
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [18.5 kB]  
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [3631 kB]  
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [648 kB]  
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 c-n-f Metadata [676 B]
```

```

Get:39 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [21.7 kB]
Get:40 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [39.6 kB]
Get:41 http://security.ubuntu.com/ubuntu jammy-security/multiverse Translation-en [8716 B]
Get:42 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 c-n-f Metadata [368 B]
Fetched 37.9 MB in 6s (5939 kB/s)
Reading package lists... Done
$ sudo apt install tree
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  tree
0 upgraded, 1 newly installed, 0 to remove and 91 not upgraded.
Need to get 47.9 kB of archives.
After this operation, 116 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 tree amd64 2.0.2-1 [47.9 kB]
Fetched 47.9 kB in 0s (2364 kB/s)
Selecting previously unselected package tree.
(Reading database ... 65857 files and directories currently installed.)
Preparing to unpack .../tree_2.0.2-1_amd64.deb ...
Unpacking tree (2.0.2-1) ...
Setting up tree (2.0.2-1) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
$ |

```

```

root@ip-172-31-84-203: ~
$ sudo apt install tree
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  tree
0 upgraded, 1 newly installed, 0 to remove and 91 not upgraded.
Need to get 47.9 kB of archives.
After this operation, 116 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 tree amd64 2.0.2-1 [47.9 kB]
Fetched 47.9 kB in 0s (2364 kB/s)
Selecting previously unselected package tree.
(Reading database ... 65857 files and directories currently installed.)
Preparing to unpack .../tree_2.0.2-1_amd64.deb ...
Unpacking tree (2.0.2-1) ...
Setting up tree (2.0.2-1) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
$ mkdir -p AWS/Devops/Cloud/New
$ tree
.
├── AWS
│   └── Devops
│       ├── Cloud
│       └── New
4 directories, 0 files
$ |

```